

Confidence Intervals

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Question 1

0/1 pt 3 19

For a confidence level of 85%, find the critical value

Question Help: [Video](#)

Question 2

0/1 pt 3 19

For a confidence level of 90%, find the critical value

Question Help: [Video](#)

Question 3

0/1 pt 3 19

Assume that a sample is used to estimate a population proportion p . Find the 99% confidence interval for a sample of size 372 with 246 successes. Enter your answer as a tri-linear inequality using decimals (not percents) accurate to three decimal places.

 $< p <$

Answer should be obtained without any preliminary rounding. However, the critical value may be rounded to 3 decimal places.

Question Help: [Video](#)

Question 4

0/1 pt 3 19

A group of researchers is interested in middle school students and their willingness to appear to stand out. A random sample of 320 middle school students is gathered, and the researchers ask them to wear bright orange shoes to school for a day. In the sample of 320 students, 249 were willing to wear the bright orange shoes. Estimate the overall proportion of middle school students that would be willing to wear bright orange shoes to school for the day. Use a 95% confidence level.

a) State the parameter of interest. Verify that the necessary conditions are present in order to carry out the procedure.

b) Find the critical value and standard error.

Critical Value: _____ Standard Error: _____

c) Find the confidence interval: (_____ , _____)

d) Interpret the 95% confidence interval in context.

Question Help:  [Video](#)

● Question 5

 0/1 pt  3  19

Part 1 of 5

In a campus-wide survey, 132 out of 157 respondents said that they approve of the way the president is handling the job. Construct a 94% confidence interval for the true percentage of the students that approve the way the president is handling the job.

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ Population standard deviation is unknown
- ☐ Simple random sample
- ☐ Normal population
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Sample size is greater than 30
- ☐ Population standard deviation is known

iii. Unknown parameter:

Question Help:  [Video](#)

● Question 6

✓ 0/1 pt ↻ 3 🔁 19

Part 1 of 5

A poll was conducted by a researcher on March 18-19, 2020, in English and Spanish, among a random national sample of 291 adults. In the poll, 218 Americans approve of the president's management of the crisis. Construct a 93% confidence interval for the proportion of the Americans that approve of the president's management of the crisis.

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ Population standard deviation is unknown
- ☐ Normal population
- ☐ Sample size is greater than 30
- ☐ Population standard deviation is known
- ☐ Simple random sample
- ☐ The number of positive and negative responses are both greater than 10

iii. Unknown parameter: Question Help:  [Video](#)

● Question 7

✓ 0/1 pt ↻ 3 🔁 19

Part 1 of 5

Alcohol abuse has been described by college presidents as the number one problem on campus, and it is an important cause of death in young adults. How common is it? A survey of 164 randomly selected U.S. college students collected information on drinking behavior and alcohol-related problems. The researchers defined "frequent binge drinking" as having five or more drinks in a row three or more times in the past two weeks. According to this definition, 137 students were classified as frequent binge drinkers. Use the data to construct and interpret a 93% confidence interval for the percentage of students who are "frequent binge drinking".

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ Population standard deviation is unknown
- ☐ Sample size is greater than 30
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Simple random sample
- ☐ Normal population
- ☐ Population standard deviation is known

iii. Unknown parameter: Question Help:  [Video](#)

● Question 8

✓ 0/1 pt ↻ 3 ⇌ 19

What is the smallest sample size that guarantees that the margin of error is less than 5% when constructing a 95% confidence interval for a population proportion?

_____ (Round UP to the nearest integer)

Question Help: [▶ Video](#)

● Question 9

✓ 0/1 pt ↻ 3 ⇌ 19

Part 1 of 5

Of 386 randomly selected medical students 22 said that they planned to work in a rural community. Find a 97% confidence interval for the percentage of all medical students who plan to work in a rural community.

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ Sample size is greater than 30
- ☐ Population standard deviation is unknown
- ☐ Normal population
- ☐ Simple random sample
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Population standard deviation is known

iii. Unknown parameter:

Question Help: [▶ Video](#)

● Question 10

✓ 0/1 pt ↻ 3 ⇌ 19

You are a researcher studying the lifespan of a certain species of bacteria. A preliminary sample of 30 bacteria reveals a sample mean of $\bar{x} = 66$ hours with a standard deviation of $s = 7$ hours. You would like to estimate the mean lifespan for this species of bacteria to within a margin of error of 0.5 hours at a 90% level of confidence.

What sample size should you gather to achieve a 0.5 hour margin of error? Round your answer *up* to the nearest whole number.

n = _____ bacteria

Question Help: [▶ Video](#)

● Question 11

✓ 0/1 pt ↻ 3 ⇌ 19

The Specific Absorption Rate (SAR) for a cell phone measures the amount of radio frequency (RF) energy absorbed by the user's body when using the handset. Every cell phone emits RF energy. Different phone models have different SAR measures. To receive certification from the Federal Communications Commission (FCC) for sale in the United States, the SAR level for a cell phone must be no more than 1 watts per kilogram. A sample of 50 models was tested and the average of their Specific Absorption Rates (SARs) was found to be 1.23 watts per kilogram. Assume that the population standard deviation is 0.25 watts per kilogram. Construct a 95% confidence interval for the average of the SARs for cell phones that received certification from FCC.

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ Sample size is greater than 30
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Simple random sample
- ☐ Population standard deviation is known
- ☐ Population standard deviation is unknown
- ☐ Normal population

iii. Unknown parameter:

Question Help:  [Video](#)

● Question 12

 0/1 pt  3  19

An incubation period is a time between when you contract a virus and when your symptoms start. By surveying randomly selected local hospitals, a researcher was able to obtain a sample of incubation periods of 16 patients, and the sample average was found to be 5.59 days. Assume that the population standard deviation is 1.7 days. Construct a 93% confidence interval for the mean incubation period of the novel coronavirus. You may also assume that the sample is from a normal population.

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ Population standard deviation is known
- ☐ Population standard deviation is unknown
- ☐ The number of positive and negative responses are both greater than 10
- ☐ Simple random sample
- ☐ Normal population
- ☐ Sample size is greater than 30

iii. Unknown parameter:

Question Help:  [Video](#)

● Question 13

 0/1 pt  3  19

One of the characteristics of a credit card is the average outstanding balance. A group of researchers wants to estimate the average outstanding balance using a randomly selected sample of 42 customers. The average monthly balance of the sample is calculated as \$1290.37. Construct and interpret a 90% confidence interval for the average outstanding balance on a credit card assuming that the population standard deviation is \$2533.

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ Simple random sample
- ☐ Sample size is greater than 30
- ☐ Population standard deviation is unknown
- ☐ Population standard deviation is known
- ☐ Normal population
- ☐ The number of positive and negative responses are both greater than 10

iii. Unknown parameter:

Question Help: [▶ Video](#)

● Question 14

✓ 0/1 pt ↻ 3 🔁 19

What is the smallest sample size that guarantees that the margin of error is less than 15 when constructing a 93% confidence interval for a population mean with the assumed population standard deviation of 303.88?

_____ (Round UP to the nearest integer)

Question Help: [▶ Video](#)

● Question 15

✓ 0/1 pt ↻ 3 🔁 19

We are interested in finding the mean water temperature of a river during the Spring run off from the nearby mountains.

You must estimate the mean temperature (in degrees Fahrenheit) with the following sample temperatures:

40.6
41.8
43.4
36.3
38.4
37.5
41.8
38.8
38.7
37.4
38

Use [StatKey](#) to get your interval.

Find the 99% confidence interval. Enter your answer as an **open-interval** (*i.e.*, parentheses) accurate to two decimal places (because the sample data are reported accurate to one decimal place).

99% C.I. =

Question Help: [▶ Video](#)

● Question 16

✓ 0/1 pt ↻ 3 🔁 19

Part 1 of 5

A common claim is that garlic lowers cholesterol levels. In a test of the effectiveness of garlic, 40 subjects were treated with doses of raw garlic, and their cholesterol levels were measured before and after the treatment. The changes in their levels of LDL cholesterol (in mg/mL) have a mean of 5.03 mg/mL and a standard deviation of 20.53 mg/mL. Construct a 90% confidence interval for the mean net change in LDL cholesterol after the garlic treatment.

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ Sample size is greater than 30
- ☐ Simple random sample
- ☐ Population standard deviation is unknown
- ☐ Normal population
- ☐ Population standard deviation is known
- ☐ The number of positive and negative responses are both greater than 10

iii. Unknown parameter:

Question Help: [▶ Video](#)

● Question 17

✓ 0/1 pt ↻ 3 ↺ 19

Part 1 of 5

For a certain drug, based on standards set by the United States Pharmacopeia (USP) - an official public standards-setting authority for all prescription and over-the-counter medicines and other health care products manufactured or sold in the United States, a standard deviation of capsule weights of less than 2 mg is acceptable. A sample of 11 capsules was taken and the weights are provided below:

100.7	99.2	99.7	99.6	99.6
99.6	100.3	102	99.5	99.7
101.4				

The mean of the data set is 100.12; the sample standard deviation is 0.891; if the normality plot is not provided you may assume that the capsule weights are normally distributed.

Construct a 95% confidence interval for the variance (standard deviation) of all capsule weights.

i. Procedure:

ii. Assumptions: (select everything that applies)

- ☐ The number of positive and negative responses are both greater than 10
- ☐ Population standard deviation is unknown
- ☐ Sample size is greater than 30
- ☐ Normal population
- ☐ Simple random sample
- ☐ Population standard deviation is known

iii. Unknown parameter:

Question Help:  [Video](#)